



REGULATIONS 2024

FLEXIBLE CHOICE BASED CREDIT SYSTEM

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

B.Tech., Artificial Intelligence and Data Science

CURRICULUM

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
1	Any 1	HS	SH1104	19EN104	Technical Writing	-	3	0	3	CGPA	Theory
2	Any 1	HS	SH1105	19EN105	Public Speaking	-	0	3	3	CGPA	Practical (10 Expts)
3	Any 1	HS	SH1101	19EN101	Communicative English	-	3	1	4	CGPA	TIP (6 Expts)
4	Exempted for LE, Any 2 for Regular	HS	SH9154	19MS154	Basic Financial Accounting	-	3	0	3	CGPA	Theory
5	Exempted for LE, Any 2 for Regular	HS	SH9156	19MS156	Human Resource Management and Team Building	-	3	0	3	CGPA	Theory
6	Exempted for LE, Any 2 for Regular	HS	SH9155	19MS155	Stock Market and Company Operations	-	3	0	3	CGPA	Theory
7	Exempted for LE	HS	SH7101	22HS101	தமிழர் மரபு/ Heritage of Tamils	-	1	0	1	CGPA	Theory (Int Only)
8	Exempted for LE	HS	SH7102	22HS102	தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	-	1	0	1	CGPA	Theory (Int Only)
9	Exempted for LE	HS	SH7801	NA	Human Values and Professional Ethics	-	2	0	2	CGPA	Theory
10		BS	SH2222	19MA222	Probability and Queueing Models	-	3	1	4	CGPA	TIP (6 Expts)
11		BS	SH2219	19MA219	Transforms and Its Applications	-	4	0	4	CGPA	Theory
12		BS	SH2220	19MA220	Mathematics for Artificial Intelligence	-	4	0	4	CGPA	Theory
13		BS	SH2221	19MA221	Linear Algebra Laboratory	-	0	2	2	CGPA	Practical (10 Expts)
14		BS	SH3214	NA	Physics for Quantum Computing	-	2	1	3	CGPA	TIP (6 Expts)
15	Exempted for LE	BS	SH4205	19CY205	Principles of Chemistry in Engineering	-	3	1	4	CGPA	TIP (6 Expts)
16	Exempted for LE	BS	SH4801	NA	Environmental Sciences and Sustainability	-	2	0	2	CGPA	Theory
17		ES	CS3301	19AI301	Python Programming	-	2	1	3	CGPA	TIP (6 Expts)
18		ES	CS3302	19AI302	Engineering Design and Modelling	-	0	3	3	CGPA	Practical (10 Expts)
19		ES	CS3303	19AI303	Engineering Mechanics and Product Development	-	2	1	3	CGPA	TIP (6 Expts)
20		ES	CS3306	19AI306	Object Oriented Programming using C++	-	2	1	3	CGPA	TIP (6 Expts)
21		ES	CS3307	19AI307	Object Oriented Programming using Java	19AI304	2	1	3	CGPA	TIP (6 Expts)

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22		ES	CS3308	19AI308	Object Oriented Programming using C#	-	2	1	3	CGPA	TIP (6 Expts)
23	Exempted for LE	ES	EC2404	19EE404	Digital Electronics	-	3	1	4	CGPA	TIP (6 Expts)
24	Exempted for LE	ES	EC2305	19EE305	Basic Electrical, Electronics and Measurement Engineering	-	2	1	3	CGPA	TIP (6 Expts)
25	Exempted for LE	ES	EC2309	19EE309	Programming Microcontrollers	-	2	1	3	CGPA	TIP (6 Expts)
26		ES	CS3304	19AI304	Fundamentals of C Programming	-	2	1	3	CGPA	TIP (6 Expts)
27		ES	CS3305	19AI305	Advanced C Programming	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
28		PC	SH2211	19MA211	Statistics and Numerical Methods	-	3	1	4	CGPA	TIP (6 Expts)
29		PC	CS1404	19CS404	Database Management System and its Applications	19AI301/19AI304/19CS301/19CS302/19CS303	3	1	4	CGPA	TIP (6 Expts)
30		PC	CS3408	19AI408	Data Structures	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
31		PC	CS3404	19AI404	Analysis of Algorithms	19AI301	2	1	3	CGPA	TIP (6 Expts)
32		PC	CS3405	19AI405	Fundamentals of Artificial Intelligence	19CS301	2	1	3	CGPA	TIP (6 Expts)
33	Exempted for LE	PC	CS3406	19AI406	Digital Image Processing Techniques	-	2	1	3	CGPA	TIP (6 Expts)
34	Exempted for LE	PC	CS1405	19CS405	Operating System	-	3	1	4	CGPA	TIP (6 Expts)
35	Exempted for LE	PC	CS1406	19CS406	Computer Networks	-	3	1	4	CGPA	TIP (6 Expts)
36		PC	CS3414	19AI414	Fundamentals of Web Application Development	-	3	2	5	CGPA	TIP (10 Expts)
37		PC	CS3407	19AI407	Parallel Computing Architecture	-	3	1	4	CGPA	TIP (6 Expts)
38		PC	CS3403	19AI403	Introduction to Data Science	19CS301	2	1	3	CGPA	TIP (6 Expts)
39		PC	CS3409	19AI409	Applied Artificial Intelligence	19AI405/19CS301	2	1	3	CGPA	TIP (6 Expts)
40		PC	CS3410	19AI410	Introduction to Machine Learning	-	2	1	3	CGPA	TIP (6 Expts)
41		PC	CS3411	19AI411	Neural Networks	-	2	1	3	CGPA	TIP (6 Expts)
42		PC	CS3412	19AI412	Web Data Mining	-	3	1	4	CGPA	TIP (6 Expts)
43		PC	CS3413	19AI413	Deep Learning and its Applications	-	2	1	3	CGPA	TIP (6 Expts)
44		PE	CS1407	19CS407	Theory of Computation	19MA206	3	0	3	CGPA	Theory
45		PE	CS1408	19CS408	Software Engineering	-	3	1	4	CGPA	TIP (6 Expts)
46		PE	CS1409	19CS409	Compiler Design	-	3	1	4	CGPA	TIP (6 Expts)

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47		PE	CS1414	19CS414	Mobile Application Development	19CS404/19CS410	1	2	3	CGPA	Theory (Int Only) and Practical (10 Expts)
48		PE	EC1408	19EC408	Microprocessor and Microcontroller	19EC303 / EC1801	3	1	4	CGPA	TIP (6 Expts)
49		PE	CS3530	19AI530	Control of Robotic Systems	-	2	1	3	CGPA	TIP (6 Expts)
50		PE	CS3531	19AI531	Cryptography and Security	-	2	1	3	CGPA	TIP (6 Expts)
51		PE	CS3532	19AI532	Information Theory for Cyber Security	-	2	1	3	CGPA	TIP (6 Expts)
52		PE	CS3533	19AI533	Introduction to Robotics	-	2	1	3	CGPA	TIP (6 Expts)
53		PE	CS3534	19AI534	Kinematics and Dynamics of Robots	-	2	1	3	CGPA	TIP (6 Expts)
54		PE	CS3504	19AI504	Mobile Robotics	-	2	1	3	CGPA	TIP (6 Expts)
55		PE	CS3505	19AI505	Reinforcement Learning	19CS301	2	1	3	CGPA	TIP (6 Expts)
56		PE	CS3535	19AI535	Robotic Sensors	-	2	1	3	CGPA	TIP (6 Expts)
57		PE	CS3536	19AI536	Steganography and Digital Watermarking	-	2	1	3	CGPA	TIP (6 Expts)
58		PE	CS3506	19AI506	Speech Analysis and Processing	-	2	1	3	CGPA	TIP (6 Expts)
59		PE	CS3507	19AI507	Special Topics in Artificial Intelligence	-	2	1	3	CGPA	TIP (6 Expts)
60		PE	CS3508	19AI508	Soft Computing Techniques	-	2	1	3	CGPA	TIP (6 Expts)
61		PE	CS3537	19AI537	Computer Graphics for Virtual Reality	-	2	1	3	CGPA	TIP (6 Expts)
62		PE	CS3538	19AI538	Mathematical Modelling and Computer Aided Engineering	-	2	1	3	CGPA	TIP (6 Expts)
63		PE	CS3509	19AI509	Concepts of Virtual And Augmented Reality	-	2	1	3	CGPA	TIP (6 Expts)
64		PE	CS3510	19AI510	Mobile VR And AI In Unity	19AI410/19CS302	2	1	3	CGPA	TIP (6 Expts)
65		PE	CS3511	19AI511	Scientific and Engineering Data Visualization	-	2	1	3	CGPA	TIP (6 Expts)
66		PE	CS3539	19AI539	Mobile User Interface Development	19CS401	2	1	3	CGPA	TIP (6 Expts)
67		PE	CS3540	19AI540	Programming Mobile Devices	19CS401	2	1	3	CGPA	TIP (6 Expts)
68		PE	CS3541	19AI541	Cloud Computing	19CS406	3	1	4	CGPA	TIP (6 Expts)
69		PE	CS3542	19AI542	Agile Software Development	19CS301	3	1	4	CGPA	TIP (6 Expts)
70		PE	CS3512	19AI512	NoSQL Database Design	19CS301	2	1	3	CGPA	TIP (6 Expts)
71		PE	CS3544	19AI544	Virtualization and Containerization	19CS405/19CS406	2	1	3	CGPA	TIP (6 Expts)

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72		PE	CS3513	19AI513	Game Programming	19CS401	2	2	4	CGPA	TIP (10 Expts)
73		PE	CS3545	19AI545	Modern Web Application Development	19CS410	2	1	3	CGPA	TIP (6 Expts)
74		PE	CS3546	19AI546	Web Server Programming	19CS401	2	1	3	CGPA	TIP (6 Expts)
75		PE	CS3514	19AI514	Self Driving Car	19CS301/19AI410	2	2	4	CGPA	TIP (10 Expts)
76		PE	CS3515	19AI515	Smart Manufacturing Technology	-	2	1	3	CGPA	TIP (6 Expts)
77		PE	CS3516	19AI516	Big Data Mining and Analysis	19AI408/ 19CS410/19IT405	2	1	3	CGPA	TIP (6 Expts)
78		PE	CS3517	19AI517	Business Analytics	-	2	1	3	CGPA	TIP (6 Expts)
79		PE	CS3518	19AI518	Cognitive Systems	-	3	0	3	CGPA	Theory
80		PE	CS3519	19AI519	Distributed Database	19CS404	3	0	3	CGPA	Theory
81		PE	CS3520	19AI520	Data Mining and Warehousing	-	2	1	3	CGPA	TIP (6 Expts)
82		PE	CS3521	19AI521	Expert Systems	-	3	0	3	CGPA	Theory
83		PE	CS3522	19AI522	Knowledge Engineering	19CS410	3	0	3	CGPA	Theory
84		PE	CS3523	19AI523	Mobile Database	-	3	0	3	CGPA	Theory
85		PE	CS3524	19AI524	Multimedia Database	19CS410	3	0	3	CGPA	Theory
86		PE	CS3525	19AI525	Video Processing	19CS301	2	1	3	CGPA	TIP (6 Expts)
87		PE	CS3547	19AI547	Blockchain for Business	19CS301	2	1	3	CGPA	TIP (6 Expts)
88		PE	CS1501	19CS501	Machine Learning	19MA212/ 19MA218	3	0	3	CGPA	Theory
89		PE	CS1511	19CS511	Natural Language Processing	19MA201	3	0	3	CGPA	Theory
90		PE	CS1519	19CS519	Data Science	19CS301/19CS501	3	0	3	CGPA	Theory
91		PE	CS1520	19CS520	Big Data Analytics	19CS411/19CS519	3	0	3	CGPA	Theory
92		PE	CS1521	19CS521	Data Warehousing and Data Mining	19CS404	3	0	3	CGPA	Theory
93		PE	CS1522	19CS522	Advanced Database Management System	19CS404	3	1	4	CGPA	TIP (6 Expts)
94		PE	CS1523	19CS523	Information Retrieval	19AI410/ 19CS501/19CS413	3	0	3	CGPA	Theory
95		PE	CS1524	19CS524	Deep Learning	19CS501	3	0	3	CGPA	Theory
96		PE	CS1526	19CS526	R Programming	19CS301/19CS303	2	1	3	CGPA	TIP (6 Expts)

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97		PE	CS1512	19CS512	Soft Computing	19CS501/19 AI410	3	0	3	CGPA	Theory
98		PE	CS1510	19CS510	Social Network Analysis	19CS501/19 CS413	3	0	3	CGPA	Theory
99		PE	CS1504	19CS504	Software Project Management	19CS408	3	0	3	CGPA	Theory
100		PE	CS1505	19CS505	Service Oriented Architecture	-	3	0	3	CGPA	Theory
101		PE	CS1503	19CS503	Developing Web Application in .Net	19CS404/19 CS410	3	0	3	CGPA	Theory
102		PE	CS1507	19CS507	Human Computer Interaction	19CS502/19 CS408	3	0	3	CGPA	Theory
103		PE	CS1502	19CS502	Computer Graphics and Multimedia	19CS401	3	0	3	CGPA	Theory
104		PE	CS1518	19CS518	Game Design and Development	19CS502	3	0	3	CGPA	Theory
105		PE	NA	19EC416	Principles of Digital Signal Processing	19EC307	3	0	3	CGPA	Theory
106		PE	EC1503	19EC503	Digital Image Processing	19EC409 / EC1409	3	0	3	CGPA	Theory
107		PE	CS1531	19CS531	Virtual Reality	19CS401/19 CS301	3	0	3	CGPA	Theory
108		PE	CS1509	19CS509	Graph Theory and Applications	19MA206	3	0	3	CGPA	Theory
109		PE	CS1517	19CS517	Mobile Computing	19CS406	3	0	3	CGPA	Theory
110		PE	NA	19EC508	Ad Hoc and Wireless Sensor Networks	19EC412	3	0	3	CGPA	Theory
111		PE	CS1508	19CS508	Parallel and Distributed Systems	19CS305/19 CS401	3	0	3	CGPA	Theory
112		PE	CS1530	19CS530	Cyber Security	19CS412	3	0	3	CGPA	Theory
113		PE	CS1514	19CS514	Network Management Systems	19CS406	3	0	3	CGPA	Theory
114		PE	CS1515	19CS515	Block Chain Architecture and its Use Cases	19CS412	3	0	3	CGPA	Theory
115		PE	CS1513	19CS513	Green Computing	-	3	0	3	CGPA	Theory
116		PE	CS1506	19CS506	Multicore Architecture	19CS305	3	0	3	CGPA	Theory
117		PE	NA	19EC414	Embedded Systems	19EC408	2	1	3	CGPA	TIP (6 Expts)
118		PE	CS1529	19CS529	Quantum Computing	19CS301/19 CS303	3	0	3	CGPA	Theory
119		PE	CS1528	19CS528	Robotics and its Applications	19CS302/19 EC408	3	0	3	CGPA	Theory
120		PE	CS1532	19CS532	Professional Readiness for Innovation, Employability and Entrepreneurship	-	0	3	3	CGPA	Practical (10 Expts)
121		PE	CS1417	19CS417	Ethical Hacking Techniques	-	2	1	3	CGPA	TIP (6 Expts)

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122		PE	CS3553	19AI553	Advanced Java Web Applications	19AI307	2	1	3	CGPA	TIP (6 Expts)
123		PE	CS1420	19CS420	Prototyping of IoT Systems	-	2	1	3	CGPA	TIP (6 Expts)
124		PE	CS4508	19AM508	Introduction to IoT	-	2	1	3	CGPA	TIP (6 Expts)
125		PE	CS1544	19CS544	Realtime Operating System	-	2	1	3	CGPA	TIP (6 Expts)
126		PE	CS1542	19CS542	Embedded Board Design	-	2	1	3	CGPA	TIP (6 Expts)
127		PE	CS1545	19CS545	Open Source Operating System	-	2	1	3	CGPA	TIP (6 Expts)
128		PE	CS1546	19CS546	Kernel Programming	-	2	1	3	CGPA	TIP (6 Expts)
129		PE	CS1418	19CS418	Cyber Law and Compliance	-	3	0	3	CGPA	Theory
130		PE	CS3543	19AI543	Software Testing	19CS301	2	1	3	CGPA	TIP (6 Expts)
131		PE	CS3548	19AI548	Robot Programming	-	2	1	3	CGPA	TIP (6 Expts)
132		PE	CS3549	19AI549	Drone Assembly and Applications	-	2	1	3	CGPA	TIP (6 Expts)
133		PE	CS3550	19AI550	Artificial Intelligence for Games	-	2	1	3	CGPA	TIP (6 Expts)
134		PE	CS3551	19AI551	Drone Design and Control	-	2	1	3	CGPA	TIP (6 Expts)
135		PE	CS3552	19AI552	Web Application User Interface Development	-	2	1	3	CGPA	TIP (6 Expts)
136		PE	CS3554	19AI554	Fundamentals of Large Language Models	-	2	1	3	CGPA	TIP (6 Expts)
137		PE	CS1558	19CS558	Data Privacy and Protection	-	2	1	3	CGPA	TIP (6 Expts)
138		PE	CS1558	19CS558	Data Privacy and Protection	-	2	1	3	CGPA	TIP (6 Expts)
139		PE	CS3555	19AI555	Exploratory Data Analysis	-	2	1	3	CGPA	TIP (6 Expts)
140		PE	CS3559	19AI559	Image and Video Analytics	-	2	1	3	CGPA	TIP (6 Expts)
141		PE	CS3557	19AI557	Neural Networks and Deep Learning	-	2	1	3	CGPA	TIP (6 Expts)
142		PE	CS3556	19AI556	Recommender Systems	-	2	1	3	CGPA	TIP (6 Expts)
143		PE	CS3558	19AI558	Text and Speech Analysis	-	2	1	3	CGPA	TIP (6 Expts)
144		PE	CS4401	19AM401	Time Series Analysis and Forecasting	-	3	1	4	CGPA	TIP (6 Expts)
145		PE	CS4501	19AM501	Machine Learning for Bioinformatics	-	3	0	3	CGPA	Theory
146		PE	CS4502	19AM502	Genetic Algorithms	-	2	1	3	CGPA	TIP (6 Expts)

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147		PE	CS4503	19AM503	Data Modeling	-	3	0	3	CGPA	Theory
148		PE	CS4504	19AM504	Brain Computer Interaction	-	3	0	3	CGPA	Theory
149		PE	CS4505	19AM505	Statistical Learning Theory	-	2	1	3	CGPA	TIP (6 Expts)
150		PE	CS4506	19AM506	Sensors and Actuators for IoT	-	2	1	3	CGPA	TIP (6 Expts)
151		PE	CS4507	19AM507	Security and Trust in IoT	-	2	1	3	CGPA	TIP (6 Expts)
152		PE	CS4509	19AM509	Industrial Internet of Things	-	2	1	3	CGPA	TIP (6 Expts)
153		PE	CS4510	19AM510	Software for Embedded Systems	-	2	1	3	CGPA	TIP (6 Expts)
154		PE	CS4511	19AM511	IoT Architecture and Protocols	-	3	0	3	CGPA	Theory
155		PE	CS4512	19AM512	Arduino for IoT	-	2	1	3	CGPA	TIP (6 Expts)
156		PE	CS4513	19AM513	Raspberry Pi for IoT	-	2	1	3	CGPA	TIP (6 Expts)
157		PE	CS4514	19AM514	Motion Planning Techniques	-	2	1	3	CGPA	TIP (6 Expts)
158		PE	CS4515	19AM515	Advanced Java Programming	-	2	1	3	CGPA	TIP (6 Expts)
159		PE	CS1558	19CS558	Data Privacy and Protection	-	2	1	3	CGPA	TIP (6 Expts)
160		PE	CS1550	19CS550	AI and Ethics	-	3	0	3	CGPA	Theory
161		PE	CS1563	19CS563	Bitcoin Mining	-	2	1	3	CGPA	TIP (6 Expts)
162		PE	CS1561	19CS561	Blockchain and Fintech	-	2	1	3	CGPA	TIP (6 Expts)
163		PE	CS3503	19AI503	Computer Vision	19CS301	2	1	3	CGPA	TIP (6 Expts)
164		PE	CS1559	19CS559	Blockchain Platforms and Its Use Cases	-	2	1	3	CGPA	TIP (6 Expts)
165		PE	CS3502	19AI502	Applied Natural Language Processing	-	2	1	3	CGPA	TIP (6 Expts)
166		PE	CS1305	19CS305	Computer Architecture	-	3	0	3	CGPA	Theory
167		PE	CS1567	19CS567	Cloud Native App Development	-	2	1	3	CGPA	TIP (6 Expts)
168		PE	CS1415	19CS415	Cryptography	-	2	1	3	CGPA	TIP (6 Expts)
169		PE	CS3526	19AI526	Information Security and Access Control	19AI527	2	1	3	CGPA	TIP (6 Expts)
170		PE	CS3527	19AI527	Security Assessment and Risk Analysis	-	3	1	4	CGPA	TIP (6 Expts)
171		PE	CS3528	19AI528	Advanced Graph Theory and Applications	19CS402	2	1	3	CGPA	TIP (6 Expts)

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172		PE	CS3529	19AI529	Advanced Data Structures	19CS401	2	1	3	CGPA	TIP (6 Expts)
173		PE	CS4516	19AM516	Python Programming for IoT	-	3	0	3	CGPA	Theory
174		PE	CS4517	19AM517	Applications of IoT in Robotics	-	3	0	3	CGPA	Theory
175		PE	CS4518	19AM518	Programming for IoT Boards	-	3	0	3	CGPA	Theory
176		PE	CS4519	19AM519	Design of Smart Cities	-	3	0	3	CGPA	Theory
177		PE	CS4520	19AM520	IoT and Multimedia Technology	-	3	0	3	CGPA	Theory
178		PE	CS4521	19AM521	Mobile Application Development for IoT	-	3	0	3	CGPA	Theory
179		PE	CS4522	19AM522	Open Source Programming for IoT	-	3	0	3	CGPA	Theory
180		PE	NA	19CS301	Problem Solving and Python Programming	-	3	1	4	CGPA	TIP (6 Expts)
181		PE	NA	19CS302	Programming in C	-	2	2	4	CGPA	TIP (10 Expts)
182		PE	NA	19CS308	Software Testing Laboratory	19CS410/19 AI401	0	2	2	CGPA	Practical (10 Expts)
183		PE	NA	19CS402	Design and Analysis of Algorithms	19CS401	3	0	3	CGPA	Theory
184		PE	NA	19CS403	Programming Paradigms	19CS401	3	1	4	CGPA	TIP (6 Expts)
185		PE	NA	19CS410	Web Programming	-	3	1	4	CGPA	TIP (6 Expts)
186		PE	CS1565	19CS565	Robotic Process Automation	-	2	1	3	CGPA	TIP (6 Expts)
187		PE	CS1566	19CS566	Software Project Management and Version Control	-	3	0	3	CGPA	Theory
188		PE	CS1579	19CS579	Generative AI Application Development	-	2	1	3	CGPA	TIP (6 Expts)
189		PE	CS1568	19CS568	Fine Tuning Large Language Models	-	3	0	3	CGPA	Theory
190		PE	CS1569	19CS569	Applied Data Science	-	2	1	3	CGPA	TIP (6 Expts)
191		PE	CS1570	19CS570	Entrepreneurship and Small Business Development	-	2	1	3	CGPA	TIP (6 Expts)
192		PE	CS1571	19CS571	Software Development and Operations	-	3	0	3	CGPA	Theory
193		PE	CS1572	19CS572	Serverless App Development	-	3	0	3	CGPA	Theory
194		PE	CS1573	19CS573	Enterprise Cloud Architecture	-	3	0	3	CGPA	Theory
195		PE	CS1574	19CS574	Advanced Class Design and Object-Oriented Techniques	-	3	0	3	CGPA	Theory
196		PE	CS1575	19CS575	Low-Code App Development	-	3	0	3	CGPA	Theory

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
197		PE	CS1577	19CS577	Cloud System Administration	-	3	0	3	CGPA	Theory
198		PE	CS1578	19CS578	Machine Learning Operations	-	3	0	3	CGPA	Theory
199		PE	NA	19CS411	Virtualization and Cloud Computing	19CS406	3	1	4	CGPA	TIP (6 Expts)
200		PE	NA	19CS412	Cryptography and Network Security	19CS406	3	1	4	CGPA	TIP (6 Expts)
201		PE	CS1416	19CS416	Cloud Security	-	2	1	3	CGPA	TIP (6 Expts)
202		PE	CS1419	19CS419	Digital Forensics and Digital Investigations	-	2	1	3	CGPA	TIP (6 Expts)
203		PE	CS1421	19CS421	Edge Computing	-	2	1	3	CGPA	TIP (6 Expts)
204		PE	CS1422	19CS422	Security for IoT Systems	-	3	0	3	CGPA	Theory
205		PE	CS1516	19CS516	Software Testing and Quality Assurance	-	3	0	3	CGPA	Theory
206		PE	CS1533	19CS533	Information Security Intelligence and Analytics using Big Data	-	2	1	3	CGPA	TIP (6 Expts)
207		PE	CS1534	19CS534	Prevention Methodology for Cybersecurity	-	2	1	3	CGPA	TIP (6 Expts)
208		PE	CS1535	19CS535	Trusted Network Systems	-	3	0	3	CGPA	Theory
209		PE	CS1536	19CS536	Firewalls and Intrusion Detection System	-	2	1	3	CGPA	TIP (6 Expts)
210		PE	CS1537	19CS537	Secure Software Engineering	-	2	1	3	CGPA	TIP (6 Expts)
211		PE	CS1538	19CS538	System Security	-	3	0	3	CGPA	Theory
212		PE	CS1539	19CS539	Hardware Security	-	3	0	3	CGPA	Theory
213		PE	CS1540	19CS540	Fog Computing	-	3	0	3	CGPA	Theory
214		PE	CS1543	19CS543	Embedded Board Manufacturing	-	2	1	3	CGPA	TIP (6 Expts)
215		PE	CS1548	19CS548	Cloud Services Management	-	2	1	3	CGPA	TIP (6 Expts)
216		PE	CS1549	19CS549	UI and UX Design	-	2	1	3	CGPA	TIP (6 Expts)
217		PE	CS1551	19CS551	Software Testing and Automation	-	2	1	3	CGPA	TIP (6 Expts)
218		PE	CS1552	19CS552	DevOps	-	2	1	3	CGPA	TIP (6 Expts)
219		PE	CS1553	19CS553	Web Application Security	-	2	1	3	CGPA	TIP (6 Expts)
220		PE	CS2401	19IT401	Information Technology Essentials	-	3	1	4	CGPA	TIP (6 Expts)
221		PE	CS2402	19IT402	Game Development Technologies	-	3	1	4	CGPA	TIP (6 Expts)

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
222		PE	CS2405	19IT405	Data Structures using Python	-	3	1	4	CGPA	TIP (6 Expts)
223		PE	CS2406	19IT406	Grid and Cloud Computing	-	3	1	4	CGPA	TIP (6 Expts)
224		PE	CS2408	19IT408	Distributed Computing	-	3	0	3	CGPA	Theory
225		PE	CS2501	19IT501	Information Theory and Coding Techniques	-	3	0	3	CGPA	Theory
226		PE	CS2502	19IT502	Enterprise Resource Planning	-	3	0	3	CGPA	Theory
227		PE	CS2503	19IT503	User Interface Design	-	3	0	3	CGPA	Theory
228		PE	CS2504	19IT504	Information Storage Management	-	3	0	3	CGPA	Theory
229		PE	CS2505	19IT505	Software Design Patterns	-	3	0	3	CGPA	Theory
230		PE	CS2506	19IT506	E-Learning Techniques	-	3	0	3	CGPA	Theory
231		PE	CS2508	19IT508	TCP/IP Design and Implementation	-	3	0	3	CGPA	Theory
232		PE	CS2509	19IT509	Advanced Algorithms	-	3	0	3	CGPA	Theory
233		PE	CS2510	19IT510	Ethical Hacking	-	3	0	3	CGPA	Theory
234		PE	CS2511	19IT511	DHTML and XML	-	3	0	3	CGPA	Theory
235		PE	CS2512	19IT512	E-Commerce	-	3	0	3	CGPA	Theory
236		PE	CS2513	19IT513	Free Open Source Software	-	3	0	3	CGPA	Theory
237		PE	CS2514	19IT514	Real Time Operating Systems	-	3	0	3	CGPA	Theory
238		PE	CS2515	19IT515	Software Agents	-	3	0	3	CGPA	Theory
239		PE	CS2516	19IT516	Business Intelligence and Analytics	-	3	0	3	CGPA	Theory
240		OE	XXXXXX	19XXXXX	Open Electives	-	0	0	0	CGPA	Theory
241		OE	XXXXXX	19XXXXX	Online Courses	-	0	0	0	CGPA	Experiential Learning
242		EEC	CS3701	19AI701	Mini Project	-	0	2	2	CGPA	Project (Int Only)
243	Option 1	EEC	CS3702	19AI702	Project Work I	-	0	2	2	CGPA	Practical (10 Expts)
244	Option 1	EEC	CS3703	19AI703	Project Work II	19AI702	0	6	6	CGPA	Project
245	Option 2	EEC	SH6802	NA	Start Up	-	0	8	8	CGPA	Project
246		EEC	SH6708	19EY708	Career Development Skills	NIL	1	0	1	CGPA	Theory (Int Only)

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
247		EEC	SH6709	19EY709	Reasoning Ability	NIL	1	0	1	CGPA	Theory (Int Only)
248		EEC	SH6710	19EY710	Quantitative Ability I	NIL	1	0	1	CGPA	Theory (Int Only)
249		EEC	SH6711	19EY711	Quantitative Ability II	SH6710 / 19EY710	1	0	1	CGPA	Theory (Int Only)
250		EEC	SH6712	19EY712	Employment Enrichment Skills	19EY702 / 19EY709 / SH6709	1	0	1	CGPA	Theory (Int Only)
251	Any 1	EEC	NA	19EY706	Company Specific Assessments For Employability	-	0	1	1	CGPA	Practical (6 Expts)
252	Any 1		SH6713	19EY713	Placement Readiness Assessment	19EY704 / 19EY711/SH 6711	1	0	1	CGPA	Theory (Int Only)
253		EEC	SH6707	19EY707	Paper Publication	-	0	0	1	NON-CGPA	Experiential Learning (Int Only)
254		MC	SH8801	NA	Constitutionalism and Artificial Intelligence Policies in India	-	1	0	1	NON-CGPA	Theory (Int Only)
255	Any 1	MC	SH8802	NA	Internship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
256	Any 1	MC	SH8803	NA	Entrepreneurship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
257	Any 1	MC	SH6803	19MC807	National Service Scheme	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
258	Any 1	MC	SH8807	19MC808	National Sports Organization	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
259	Any 1	MC	SH8808	19MC809	Youth Red Cross	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)

BATCH: 2024- 2028 Onwards		
Category	Credits to be Earned (Regular)	Credits to be Earned (Lateral Entry)
Humanities and Science	14	3
Basic Science	23	16
Engineering Science	28	15
Professional Core	56	48
Professional Electives	16	15
Open Electives	12	12
Employability Enhancement Courses	16	16
Mandatory Courses	4	4
TOTAL	169	129